TORSION PRIMES IN DEGREE 10 -- AUDITED VERSION
28 October 2026

Result:

    S(10) = {2,3,5,7,11,13,17,19,23,29,31,37} = Primes(37).

Here S(10) means degree exactly 10, with arbitrary elliptic curves over the number field. It does not restrict the elliptic curve to being defined over Q.

This version uses the fresh full-curve certificates in work/audit11.
The earlier nonic and decic local exclusion results are not hypotheses.
The detailed construction and independent verification of those certificates are in S11_proof_audited.pdf. The present note checks their applicability to degree 10, together with its separate prime bound and occurrences.
No asymptotic conjecture in Derickx--Stoll is assumed.

1. Occurrence

S(5)=Primes(19), so quadratic base extension gives every prime at most 19.
Fresh Magma calculations verify degree-exactly-10 examples for 23,29,31,37.
The order-37 point is an entry in the local LowDegreePlaces table that was missed in the initial investigation. In particular S(10)=Primes(31) is false.

Here is a uniform description of the examples for 29,31,37. Let a be a root of f, x=a, y=g(a), and set

    r=(x^2*y-x*y+y-1)/(x^2*y-x), s=(x*y-y+1)/(x*y),
    b=r*s*(r-1), c=s*(r-1).

Then P=(0,0) on

    E: Y^2+(1-c)XY-bY = X^3-bX^2

has the asserted prime order. Irreducibility of f and the identity pP=O were checked exactly, as were nonsingularity and P!=O.

p=29:
    f=t^10-3t^9+8t^7-5t^6-6t^5+5t^4-2t^2+1,
    g=(2t^9+t^8-11t^7-8t^6+49t^5-77t^3+35t^2+46t-13)/29.
p=31:
    f=t^10-4t^9+3t^8+6t^7-2t^6-8t^5-8t^4+11t^3+6t^2-5t+1,
    g=(109t^9-266t^8-207t^7+801t^6+606t^5-304t^4
       -1470t^3-410t^2+1000t+71)/215.
    Here j(E)=0.
p=37:
    f=t^10-5t^8+3t^7+10t^6-8t^5-7t^4+9t^3-3t+1,
    g=2t^9+t^8-10t^7+t^6+23t^5-6t^4-22t^3+11t^2+9t-5.
    Here j(E) has degree 10.

For 23, take a root a of the irreducible polynomial
    f=t^10-3t^9+(3/4)t^8+(7/2)t^7+(13/8)t^6-(17/2)t^5
      +(87/16)t^4-(7/8)t^3+(5/32)t^2-(3/16)t+1/32.
The curve with Weierstrass invariants [A,B,B,0,0], where
    A=-8a^3+24a^2-22a+7,
    B=32a^5-144a^4+240a^3-180a^2+58a-6,
has (0,0) of order 23. This comes from specializing the degree-10 function 1/(x*y) on the local model of X_1(23) to 2.

2. A finite upper bound: no prime greater than 101

Use Derickx--Stoll, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1, Remark 4.4, Proposition 5.2 and Corollary 7.5.

Oesterle gives p<=(3^5+1)^2=59536. For p>19 any modular point arising in degree 10 has exact degree 10: a smaller residue degree must divide 10, and S(1), S(2), S(5) contain no p>19.

For every prime 131<=p<=59536, Corollary 7.5 applies beyond its small-prime alternative because p^2>(2^19/325)*10+1. Its two exceptional CM possibilities would give p=21 or p=31. Otherwise it requires

$$(p-1)/(2*\mathrm{str}(p)) \mid 10,$$

where str(p) is the lcm of the orders of strange characters and is 1 if there are none. The complete strange-character table makes this condition false for all 5987 primes in the indicated range. The script verify_bound.py checks the arithmetic against the saved table.

The remaining primes above 101 are 103,107,109,113,127. They are not strange, and 325*(p^2-1)/65536>50. Proposition 4.1 with a=2, ell=2, n=5 therefore makes the trace divisor invariant under <2>. Its orders in Delta(p) are respectively 51,53,18,14,7. The possible geometric orbit lengths (including elliptic-point ramification by 2 or 3) never divide 10. An irreducible degree-10 point is therefore impossible. Remark 4.4 supplies the rational 2-torsion hypothesis.

3. FRESH LOCAL CERTIFICATES IN DEGREE 10

For p>37, a point arising over a degree-10 field has residue degree exactly 10: smaller residue degrees divide 10 and are excluded by the known S(1), S(2),S(5). Let D be its irreducible trace divisor of degree 10.

The audit for S(11) recomputed the complete finite-field census through degree 11 and Hecke obstructions on the FULL modular curves for every prime 37<=p<=103. These calculations, rather than the conclusion about S(11), apply here. Their dependence on the total degree is checked below.

3a. Rational-cusp reductions

The fresh degree-11 endomorphism certificates give the degree-10 formal immersion criterion as well. Normalize a degree-10 cusp multiplicity tuple so that its first entry is maximal, and add one to that entry. This is a degree-11 tuple included in the exhaustive successful check. The original ten Hecke operators are a subset of the eleven independent operators in that check, so they are independent. The endomorphisms used annihilate the rational Jacobian group independently of the degree. Thus the same formal immersion argument excludes every reduction of D supported on rational cusps. This uses the new index-20 combined map at 41 and the other fresh maps specified in the S(11) audit.

3b. Ordinary noncuspidal reductions

At p>37 a degree-10 reduction has at most one ordinary noncuspidal closed point P. The minimum noncuspidal degree is at least 6 except at 41, where all degree-5 points are supersingular, there are no degree-6 or degree-7 points, and the minimum ordinary degree is 8. Its complementary cusps are rational. The fresh ordinary orbit counts through degree 10 are:

p	41	43	47	53	59	61	67	71	73	79	83	89	97	101
orbits	13	11	5	8	8	7	8	7	4	2	9	5	4	5

All 96 orbits have non-principal alpha*A_3*P, certified by exact canonical-ring multiplication and independently checked full matrix ranks. The endomorphisms alpha are listed in Section 5 of the S(11) proof, including the stronger operators used at 97 and 101.

The obstruction applies to D of degree 10 without alteration. The class alpha[D-10c] is rational torsion; A_3 kills it, and A_3 kills the rational cusp complement. Therefore alpha*A_3*P would be principal. Its verified non-principality is a contradiction. The finite-image assertion for alpha, the target trace multiplicities, and the divisor obstruction do not use that the total degree is 11. In particular the former quotient-lifting issues at 97 and 101 are absent from this proof.

3c. Supersingular noncuspidal reductions

The fresh census leaves three supersingular orbit types for p>37. At 41, degree 5, the native A_3 non-principality checks exclude a single closed point and every unordered pair, including repetition. These are the only possible multiple-noncuspidal-point reductions in degree 10. At 43, degree 7, the native alpha*A_3 check excludes the entire diamond orbit. Both calculations remain obstructions after adding rational cusps.

At 73, the supersingular points have degree 6 and are fixed as closed points by h=<2>, of order 9. Let C be the rational lift of the complementary cusp divisor, now of degree 4. The degree-zero class y=[h(D-C)-(D-C)] is rational torsion and reduces to zero. Hence 2y=0. The fresh exact JOne homology certificate shows that h acts trivially on rational 2-torsion, so the norm relation also gives 9y=0 and y=0. The gonality bound is greater than 26>10+4. It follows that hD+C=D+hC, hence hC=C. The free order-9 action on rational cusps would require 9|4, a contradiction.

Here is the torsion justification when Z=D-C does not have degree zero. If h has order m, put N_h Z=sum_(i=0)^(m-1) h^i Z. The divisor mZ-N_h Z has degree zero, and m[hZ-Z]=(h-1)[mZ-N_h Z]. Finiteness of (h-1)J(Q) therefore makes m[hZ-Z], and hence [hZ-Z] itself, torsion. This is the precise degree-zero interpretation used in all these stabilizer arguments.

3d. Nonrational cusp reductions

Only p=41,43,73 have nonrational closed cusps of degree at most 10. At 41, either degree-10 cusp has non-principal A_3 image by the fresh native computation. Since J_1(41)(Q) is finite, this is impossible for D.

At 43, a degree-7 cusp has a rational-cusp complement C of degree 3. The diamond h=<2>, of order 7, fixes that closed cusp. The fresh integral homology certificate shows trivial h-action on rational 2-torsion. As above, y=[h(D-C)-(D-C)] is killed by 2 and by the odd norm 7, so y=0. The NEW proved bound gon_Q X_1(43)>=16 is greater than 10+3=13. Thus hD+C=D+hC, which would require 7|3. This supplies the gonality bound needed here without assuming the stronger website bound quoted in the earlier writeup.

At 73, the nonrational cusps have degree 9, with rational complement C of degree 1. The order-9 diamond h=<2> fixes the closed cusp; its action on rational 2-torsion is trivial. The norm and gonality arguments would require 9|1. There are no other cuspidal cases.

These cases are exhaustive, so every prime p>37 is excluded. Together with the separate prime bound and occurrence checks in Sections 1 and 2, this proves S(10)=Primes(37).

4. AUDIT STATUS AND REPRODUCTION

The global bound and the degree-10 occurrences for 23,29,31,37 were rerun when preparing this version. Occurrence through 19 uses the published S(5) theorem and quadratic extension. The raw local certificates are the fresh computations described in S11_proof_audited.pdf, with the degree-10 transfer established in Section 3 above. This proof does not assume the old quotient, lifting or nonic exclusion certificates.

The case ledger and its closure checker are:

- work/audit10/final_result.json
- work/audit10/certify_audit10.py

The complete local certificate archive is S11_audited_certificates.zip. The mathematical references are those listed in the S(11) proof, together with the published S(5) classification. The separate S(10) bound needs only the published strange-character table below 100000, since Oesterle’s degree-10 cap is 59536.

Both proofs are computer-assisted arguments checked in this workspace. A subsequent independent AI-agent review checked their mathematics and certificates and found no blocking gap. This is not external human peer review or formal verification in a proof assistant. The review report and the recorded checks specify the basis for these conclusions.